

# DAX RAJANI

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## WORK EXPERIENCE

**Glide Technology Pvt. Ltd.** | Ahmedabad, India

Jul 2023 – Aug 2024

*Associate Engineer, Embedded Firmware*

- Designed and implemented a complete MiWi proprietary protocol connection management module in embedded C on ARM Cortex-M under FreeRTOS, handling frequency channel scanning, hub registration validation, device identifier exchange, and automatic broadcast-mode fallback to sustain uptime across a **4,000+ node** real-time production deployment.
- Optimised real-time power management firmware by migrating the BLE stack from Bluetooth Classic to BLE and restructuring FreeRTOS task scheduling, reducing device power consumption by **30%** while maintaining continuous core temperature monitoring across all sensor nodes.
- Improved WiRa indoor positioning accuracy by **25%** by implementing RSSI-based distance calculation with data-frame validation filters that discarded non-compliant BLE advertising packets at the driver level in real time.
- Designed secure embedded C communication protocols including BLE data parsing, CRC validation, and interrupt-driven retry logic between host and slave ARM Cortex-M controllers over UART and SPI, increasing packet transfer reliability by **20%**.
- Produced firmware design documents, test plans, and release notes; conducted peer code reviews; and supported system bring-up and integration testing in collaboration with hardware and systems engineering teams.

**Glide Technology Pvt. Ltd.** | Ahmedabad, India

Dec 2022 – Jun 2023

*Software Engineering Intern, Embedded Systems*

- Resolved a critical memory management defect in production embedded C firmware within **2 weeks** by tracing heap allocations and deallocation sequences across a function-pointer-based state machine codebase, restoring stable operation to **4,000+ deployed sensors** resetting every 5–10 seconds due to memory exhaustion.
- Prototyped a WiRa positioning system on the DA14695 microcontroller in embedded C with Yocto Linux, configuring BLE advertising, RSSI-based distance calculation, and data-frame validation logic for filtering malformed sensor packets before driver-level processing.
- Built practical foundation in bare-metal C, FreeRTOS, memory management, interrupt handling, Yocto Linux, I2C, SPI, UART, cross-compiler toolchains, and hardware debugging with JTAG, oscilloscopes, and logic analyzers.

## PROJECTS

**Health Symptom Analyzer** | Python, PyTorch, scikit-learn, XGBoost, NumPy, Streamlit

Fall 2025

- Built an end-to-end ML pipeline using a **7-model Soft Voting Ensemble** (Random Forest, SVM, Naive Bayes, KNN, Decision Tree, Logistic Regression, XGBoost) to classify **41 disease classes** from patient symptoms, deployed as a real-time Streamlit web app with sub-**50ms** inference, model explainability, and automated evaluation metrics.

**E-Commerce Analytics Pipeline** | Apache Spark, GCP Dataproc, Python, PySpark

Winter 2026

- Designed a distributed analytics pipeline on GCP Dataproc using Apache Spark, implementing sessionisation, funnel analysis, last-touch attribution, and anomaly detection on a **42M row dataset** across a 9-configuration cloud scaling study (3 cluster sizes and 3 data sizes).

## EDUCATION

**Concordia University** | Montreal, QC

2024 – 2026 (Expected)

*Master of Engineering, Electrical and Computer Engineering*

**Relevant Coursework:** Applied Machine Learning, Computer Neural Networks, Biological and Medical Image Processing, Distributed Systems, Embedded Systems

**Ganpat University** | Gujarat, India

2019 – 2023

*Bachelor of Technology, Computer Engineering*

## ACHIEVEMENTS

- Awarded **Emerging Star of the Year 2023–24** at Glide Technology for technical contributions across multiple embedded firmware production deployments.
- Recognized as Volunteer of the Year 2023–24 for organising corporate events and managing digital content production.

- Earned 3 provincial and district gold medals in Powerlifting, demonstrating sustained high-performance discipline and competitive resilience.

## KEY SKILLS

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**Embedded Firmware:** C, Embedded C, C++, Python, Bare-Metal, State Machine Design, Memory Management, Interrupt Handling, Multithreading, Modular Architecture

**Microcontrollers and Hardware:** ARM Cortex-M3/M4, Dialog DA14695, Microchip Studio, NXP, JTAG, Logic Analyzer, Oscilloscope, Bit Manipulation, Memory-Mapped I/O

**RTOS and OS:** FreeRTOS, Proprietary RTOS, Yocto Linux, Embedded Linux, Cross-Compiler Toolchains, Task Scheduling, Power Optimisation

**Protocols:** BLE, MiWi Proprietary, UART, I2C, SPI, USB, CRC Validation, Serial Communication, Driver Development

**AI and ML:** PyTorch, scikit-learn, XGBoost, NumPy, Matplotlib, Ensemble Methods, Model Optimisation, Applied ML, Neural Networks

**Tools and SDLC:** Git, GCP Dataproc, Apache Spark, Streamlit, Agile Scrum, Unit Testing, Integration Testing, Code Reviews, Technical Documentation